



DEPARTMENT OF ENGINEERING MATHEMATICS

Multi-Granular Forecasting of Emergency Department Demand: Replication and Post-2022 Extension

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Supervisor: Dalila O'Grady

Declaration

This dissertation is submitted to the University of Bristol in accordance with the requirements of the degree of MSc in the Faculty of Engineering. It has not been submitted for any other degree or diploma of any examining body. Except where specifically acknowledged, it is all the work of the Author.

Cathy Mariam Vijay, Friday 4th September, 2026

Ethics Statement

This project did not require ethical review, as determined by my supervisor, Dr. Dalila O'Grady. I have attained a score of 12/12 on the ethics assessment.

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Abstract

The number of patients visiting the emergency departments in England has continued to rise throughout the last ten years; however, forecasting models have not been evaluated outside their training phases and geographical settings. In this study, we extend the Multi-Granular Stacked Regression (MGSR) model suggested by James, Wood and Denholm (2023) [11], which uses monthly service capacity and annual population health characteristics to predict the number of visits to Emergency Departments at the CCG level.

The replication, built independently from raw NHS and ONS sources for 2018-2019, reproduces the capacity submodel ($R^2=0.42$ vs. reported 0.41) and confirms the stacking architecture’s complementary-information hypothesis, though the population health submodel proves sensitive to small data perturbations ($R^2=0.54$ vs. 0.62). The combined result using the MGSR architecture of the replication achieves similar conclusions ($R^2 = 0.76$ vs $R^2 = 0.79$) to the original paper, with the stack having improved on the individual models.

The extension applies the MGSR to the post-2022 period, navigating three structural discontinuities: the replacement of 200+ CCGs with 42 Integrated Care Boards, the discontinuation of the ONS Health Index, and the loss of Sub-ICB resolution in 111 and ambulance data. Using Hospital Episode Statistics emergency admissions as a genuinely disaggregated target across 97 Sub-ICB Locations, with QOF chronic disease prevalence indicators replacing the Health Index, and with tuned Gradient Boosting Level-0 learners, the stacking implementation reaches cross-validated $R^2 = 0.88$ consistent with, but not decisive evidence for, the complementary-information property and recovers Level-1 weights of 0.27 for the capacity block and 0.84 for population health under the extended specification (0.13 and 0.98 under the capacity block as specified), close to the published 0.22 and 0.85.

Nevertheless, a flat single-model specification on the same features slightly outperforms the stack ($R^2 = 0.91$ vs. 0.88, $p = 0.0001$), while even the group-mean baseline predicting each Sub-ICB by the corresponding historical mean reaches $R^2 = 0.81$. This means that most of the explained variance is the area-identity effect rather than a generalisable relationship. When geography-wise holdout is applied, the model deteriorates to $R^2 = 0.08$ for the test Sub-ICBs and underperforms the naive baseline for the test ICBs. In the case of temporal holdout, when a specification including the one month demand lag is added, the model predicts 2024 demand with $R^2 = 0.86$ and MAPE = 7.2%, beating persistence, seasonal naive, SARIMA and exponential smoothing baselines. However, this study finds that almost all of the gain is due to the lag term.

Supporting Technologies

The following third-party resources were used during this project:

- Python 3 was used as the primary programming language for all data processing and modelling.
- The `scikit-learn` library was used for Random Forest, Gradient Boosting, cross-validation, and all model evaluation.
- `XGBoost` was used as an alternative gradient boosting implementation at Level-0.
- `pandas` and `NumPy` were used for data manipulation and numerical computation.
- `matplotlib` and `seaborn` were used for all figures and visualisations.
- The `fingertips_py` API client was used to retrieve QOF chronic disease prevalence indicators from the OHID Fingertips platform.
- `statsmodels` was used for SARIMA and exponential smoothing baseline models.
- NHS Digital open data publications (HES Monthly Activity Report, Appointments in General Practice, AmbSYS Time Series, NHS 111 Minimum Dataset) and ONS mid-year population estimates were used as primary data sources.
- $\text{\LaTeX}$ was used to prepare this dissertation, via the online service Overleaf.

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Chapter 1

Background

1.1 The Problem of Forecasting Emergency Department Demand

Attendances at emergency departments in the NHS have been rising steadily over the last decade and unplanned demand of this kind is one of the main pressures on hospital capacity planning [14, 24]. Accurate short- and medium-term forecasts enable commissioners and trusts to plan staffing, bed capacity and ambulance and primary-care resourcing ahead of demand spikes, rather than reacting to them once pressure has already materialised. Slow-moving, structural population health characteristics such as age profile, deprivation and chronic disease prevalence, define a baseline level of expected demand for a given area, one which changes only slowly from year to year [1]. The burden of chronic disease is a particularly strong driver: population-based evidence shows that multi-morbidity is associated with substantially higher rates of ED attendance and emergency admission [9, 28]. Fast-moving operational capacity signals such as availability of GP appointments, NHS 111 call volumes, ambulance response activity, modulate that baseline month to month as pressure is displaced between different parts of the urgent and emergency care system [6]. A model built on one signal will either misestimate the baseline level of demand in an area or will miss the short term variations around it.

1.2 The Multi-Granular Stacked Regression Approach

James et al. [11] overcome this limitation by modelling the two types of signals into one set of features and suggesting a Multi-Granular Stacked Regression (MGSR) that treats the two signals in terms of their respective granularity and combines them only in the final stage. The approach consists of a stack containing two levels using Random Forest models, Level 0a which includes the capacity submodel and Level 0b which includes annually varying population health features. The first level (Level 0a) takes as input monthly varying operational features (GP appointment availability, NHS 111 calls offered, ambulance calls answered), predicting ED attendances directly at CCG-month granularity. In the second submodel at level 0 (Level 0b) the paper uses annually varying population health features at CCG-year granularity; modeling it at CCG-month granularity would artificially increase the sample size because features that do not change within a year would appear in 12 rows. Level 1 then combines the two level 0 predictions using Ordinary Least Squares regression using a set of learned weights to collate the outputs into a unified prediction.

The reported figures make the case for the architecture on its own terms: R^2 of 0.41 for the capacity submodel in isolation, 0.62 for the population health submodel in isolation, and 0.79 for the combined stack, evaluated by repeated 5-fold cross-validation on 2018–2019 CCG-level data. Given the combined model substantially outperforms either submodel alone, it is presented as evidence that capacity and population health features carry complementary, non-redundant information about ED demand. The improvement indicates that the stack recovers signals from the data that neither submodel captures on its own, making the multi-granular design the central methodological contribution of this paper and the primary component evaluated in the replication of this project.

1.3 Ensemble and Stacked Regression Methods

Random Forests [3] are a popular ensemble method for tabular regression, and are the default choice at this scale due to their tolerance to feature scaling and low need for hyperparameter tuning. This technique, where the predictions of many base learners are stacked or merged, is a well-known ensemble technique that improves generalisation by learning how to optimally weight and integrate the outputs of diverse base learners [27]. In the MGSR architecture, stacking serves a purpose other than variance reduction as it is the mechanism by which two feature sets of completely different granularity are reconciled into a single prediction. As Level-0 models are trained on different row groupings of the same underlying data before their outputs are combined at Level 1.

Fitting the population health submodel on the entire dataset before passing its predictions to the Level-1 combiner is a recognised leakage risk as the Level-0 predictions already “know” the training-set target, which runs the risk of inflating the reported R^2 relative to a held-out estimate. The method that was published and this project’s replication of it avoids this by rebuilding the population health submodel from scratch inside each cross-validation fold, so that predictions on held-out data are generated without either the meta-model or the population health submodel seeing the target values at any point before.

Furthermore, boosted tree approaches, like Gradient Boosting [7] and XGBoost [4], are commonly claimed to beat Random Forests when it comes to structured, tabular prediction tasks in domains that are comparable due to their sequential error-correction mechanism. This drives their use in the current project as possible replacements for Level-0 Random Forests, which are tested directly against the published specification throughout the extension stage.

1.4 Validation Strategy for Time-Structured Data

In this dataset, the data is constructed through monthly observations by geography as it contains both a cross-sectional and a temporal component, and the choice of validation scheme determines what a reported R^2 is actually measuring. The standard k -fold cross-validation used in the original paper and reproduced in the replication stage of this project, shuffles rows across folds regardless of time. Unlike typical forecasting validation, this test assesses interpolation within the training period, and importantly, the model is permitted to train on data points that temporally follow the test set, which effectively allows it to access information from later time periods.

The two alternatives that avoid this issue are blocked or rolling-origin temporal validation and grouped cross-validation. The blocked temporal validation always trains on an earlier block of time and evaluates on a later one, which is the appropriate scheme for a model designed to forecast forward rather than interpolate within an observed period [10]. Alternatively, grouped cross-validation, in which whole geographic units are withheld, tests whether the fitted relationship transfers to an area the model has never observed. This is significant for panel data in which many features are constant or near-constant within a unit, because a model may recover the unit’s average level without learning any relationship that generalises beyond it. This project therefore reports three schemes for its post-2022 extension: repeated k -fold, GroupKFold holding out whole Sub-ICBs and whole ICBs, and a fixed temporal train/test split (Section 4.4). Scores under the stricter schemes are typically lower, and the gap between them is a property of the validation design rather than evidence about the model.

When evaluating forecast, we require quality baseline methods that are appropriate to the task. Persistence and seasonal-naïve are standard baselines in the forecasting literature [10], the former predicting that next period equals this period and the latter that next period equals the same period one year prior. Importantly, any machine learning model’s added value should be demonstrated against both at each forecast horizon, without which, a model reporting a high R^2 may simply be absorbing the strong autocorrelation already present in the target series.

1.5 NHS Structural Context: From CCG to ICB, and the Discontinued Health Index

The existing MGSR dataset spans 2018-2019, where the NHS in England had its organisational structure based on Clinical Commissioning Groups. As of July 2022, however, CCGs were abolished, with Integrated Care Boards taking their place as fewer but larger geographical divisions [15], while retaining Sub-ICB Location as an additional layer within each ICB. Therefore, a new dataset constructed from post-2022

data sources cannot simply be seen as a continuation of previous CCG-level dataset, since there have been changes to the geographical boundaries, coding systems and the levels of disaggregation for various data sources, while ambulance activity and 111 call data are still available only on a less fine geographical level.

A second independent feature discontinuity related to the population health metrics, in which the Composite Health Index, where the People/Places/Lives feature set was based, was discontinued after 2021 [16], thus leaving no suitable substitute for the original population health features after 2022. This study provides an alternative through the age structure features (population and aged 65+ count and percentage), derived from ONS mid-year population estimates, because age structure is one of the more straightforward proxies for baseline health care demand, although this is a less comprehensive feature set than the original Health Index.

Geographic reorganisation and the loss of the Health Index, together, explain why the post-COVID period is regarded here as a true methodological extension. Both factors influence the data engineering and modelling decisions described in the following chapter, with the former motivating the shift to Sub-ICB-level modelling and the latter requiring the replacement of OHID Fingertips indicators. These changes are evaluated alongside the orthogonal-signal stacking architecture as part of the extension's overall methodological design.

1.6 Research Questions

The preceding sections identify a model whose central claim rests on evidence from a single dataset, evaluated under a validation scheme that does not distinguish interpolation from transfer, in a data environment that no longer exists. This study accordingly asks four questions; the first is addressed by the replication in Section 3; the remaining three by the extension in Section 4.

- RQ1. Does the complementary-information hypothesis underlying the MGSR hold outside its original setting?** That is, does a stack combining separately modelled capacity and population health blocks outperform each block in isolation once the geography, the dependent variable, and the population health feature set have all changed?
- RQ2. Is the multi-granular separation necessary for that performance?** A single flat model given both feature blocks simultaneously provides the relevant comparison, and if it matches the stack, the architecture's justification must be sought in properties other than predictive accuracy.
- RQ3. What does the cross-validated R^2 conventionally reported for models of this kind actually measure?** Where features are constant or near-constant within a commissioning area, a model may recover the identity of the area rather than a relationship that transfers to a new one. It is important to know the extent to which the reported accuracy changes when whole Sub-ICBs and whole ICBs are withheld and how much of it exceeds a baseline that predicts each area by its own historical mean.
- RQ4. Does the multi-granular framing remain the appropriate one once demand history is available?** The original architecture is motivated by settings in which no such history exists and where a lagged target can be constructed. The relevant question is whether structural features add forecasting information beyond it.

Section 5 returns to each in turn.

Chapter 2

Literature Review - Supporting Papers

This chapter reviews four papers that place the MGSR in the wider ED forecasting literature. Each follows the register of the existing James et al. [11] review: extraction of the paper’s design and findings, then critical engagement with the relationship between its claims and its evidence. Together they cover the main method families in the field: classical time series, machine learning, deep learning, and econometric multivariate models. They set the gap the present study addresses.

2.1 Cheng et al. (2021)

Cheng et al. [5] use a full year of ED visit records from a large academic centre, they forecast hourly ED occupancy up to four hours ahead, fitting a separate seasonal autoregressive integrated moving average model with external regressors (SARIMAX) for each hour of the day.

The structure of the data has an hour-specific specification where ED occupancy exhibits both time-of-day and day-of-week periodicity, and a single pooled model combines these two seasonalities into a functional form. The authors find SARIMAX delivers the most accurate hourly forecasts, benchmarking against a rolling average, plain ARIMA, vector autoregression, and Holt-Winters exponential smoothing. It is evaluated through mean squared error, mean absolute error and MAPE alongside coverage of 80% and 95% prediction intervals.

The paper’s principal value to this review is as a demonstration of what the ARIMA family can achieve under the conditions in which differencing and seasonal terms are well specified. That is, if the target is a short-horizon, high-frequency, univariate-dominant series with strong and stable periodicity. The paper finds that SARIMAX outperforms VAR, which is important, since VAR is the model [17] later found to be the most effective for daily boarding volumes. Overall, the relative ranking of econometric methods depends on the specific resolution and target under consideration.

There are two main limitations when comparing this study to the Multi-Granular Statistical Regression (MGSR) model. First, this study uses a highly granular model with a four-hour forecasting horizon. Second, the SARIMAX architecture cannot perform the range of tasks addressed by the MGSR. This level of granularity is useful for staffing decisions within a shift but does not apply to the strategic planning horizon or the broader scope of the MGSR. Therefore, the two models address different research questions, and their accuracy metrics are not directly comparable.

Since ARIMAX requires knowledge of the values that the exogenous regressors will take over the forecast period, it is a straightforward demand when it comes to calendar variables over a four-hour period, but not when carrying out a multi-year forecast based on population health and service capacity. Additionally, ARIMAX introduces another element of complexity and difficulty in that it is necessary to have a second model for forecasting the exogenous regressors before feeding those forecasts into the ARIMAX model. James et al. [11] themselves identify this constraint as their reason for excluding ARIMAX from consideration, and Cheng et al. [5] illustrate why that reasoning holds and thus establish the strength of the classical approach in its domain and the border beyond which it is not possible to extend it.

2.2 Sudarshan et al. (2021)

Sudarshan et al. [23] compare deep learning architectures directly against a tree ensemble, which James et al. [11] cite when conceding that neural networks outperform linear models in certain settings. To forecast daily ED patient arrivals, the authors evaluate a Random Forest regressor against a convolutional neural network and a long short-term memory network, incorporating historical arrivals with meteorological and calendar variables. The performance is separated by the forecasting horizon where CNN achieves the lowest error for forecasting a 3-day horizon (MAPE 9.24%), while LSTM performs best for the same-day horizon (MAPE 8.04%) and the 7-day horizon prediction (MAPE 8.91%). Both of these deep models outperform the Random Forest baseline.

The horizon-dependence of the ranking is the most useful finding for present purposes, and it is more consequential than the headline claim that deep learning is better. Different architectures perform best at different horizons, suggesting that no single model is consistently optimal across the forecasting problem. The authors do not draw this inference, and the study contains no ensemble condition against which the individual models could be compared.

Three limitations determine how far the result can be carried: First, the performance margins are narrow and the stability of the ranking is therefore unclear, since the spread between the best and worst reported MAPE is less than two percentage points, with no significance testing reported across forecast horizons. The reported ranking may simply be a byproduct of a particular train-test partition. Second, the single Random Forest model tested was not a strong representative of the method’s capability. This dissertation’s own replication demonstrates how Random Forest performance varies with ensemble size. A single-configuration baseline is weak evidence that tree methods are inherently inferior to deep architectures.

Third, and most relevant to the MGSr, the deep models are trained on daily arrivals with weather and calendar covariates over a single site. LSTM architectures require long sequences to learn temporal dependencies, and while the daily observations over several years at one hospital provide this sequence length, the monthly observations over a two-year study period do not, making deep learning an unviable extension for MGSr at its published resolution. This also places a constraint on the dissertation’s post-pandemic extension, in that, across 2022–2024, there are only 33 monthly observations per Sub-ICB, which remains a relatively small sample for these models. Extending the period beyond 2024 would therefore provide a stronger basis for using such architectures.

2.3 Vollmer et al. (2021)

The most complete comparison of methods in this analysis is provided by Vollmer et al. [25], which finds that the belief that the more complex designs improve accuracy in forecasting is mistaken. They examine four classical time series methods, including ARIMA, exponential smoothing, seasonal decomposition of time series by LOESS, and structural time series models, using eight years of daily attendance data from St Mary’s and Charing Cross hospitals in London. They then compare these methods with four machine learning techniques, namely penalised regression using glmnet, random forests, gradient boosting machines, and k-nearest neighbours. Finally, they test a linear stacked regression model that combines all eight methods.

The models forecast attendance at one, three, and seven day horizons using predictors including lagged attendance, bank and school holidays, precipitation, temperature, Google Trends search volume for ‘flu’, and locally known events such as the Notting Hill Carnival. This feature set allows the study to test whether more complex models can extract useful information beyond historical demand. However, simple linear models perform as well as or better than the tree ensembles throughout, with the best one-day forecasts achieving a mean absolute error of approximately 14 patients at St Mary’s (MAPE 6.8%) and 10 at Charing Cross (MAPE 8.6%). These figures come from ordinary linear regression and glmnet as opposed to Random Forest or gradient boosting, with the stacked regression achieving MAEs of 14.51 and 10.60 at the two hospitals, respectively. The authors argue that the benefit of stacking comes from protecting against the failure of any single model, with the stacked model being slightly worse than the best individual model but never significantly worse than any of them.

This finding requires contextualisation before it can be generalised, because it is a property of the predictor set rather than of the algorithms. The paper’s own variable importance analysis establishes that the dominant predictor across all four machine learning models is average demand over the preceding week, followed by day-of-week and month indicators. This is autoregressive or temporal data which is smooth

and predictable. Since it is a smooth trend, there is little nonlinear interaction structure available for tree methods to exploit, and a linear model with appropriate lag terms and calendar dummies will capture most of the recoverable variance. The MGSr’s Population Health model presents the opposite case. It has heterogeneous, slowly varying, cross-sectional variables like the Health Index sub-domains, population size, age distribution, whose interactions are precisely the type of structure these tree ensembles are built to detect. The present study’s finding that gradient boosting raises Population Health R2 from 0.54 to 0.70 is therefore not in conflict with Vollmer et al. [25], but delimits their conclusion: linear methods suffice under autoregressive feature regimes, and tree ensembles earn their complexity only where the feature space is heterogeneous and cross-sectional.

A second point relates to comparability, as Vollmer et al. [25] forecast daily attendances at individual hospitals averaging 208 and 106 patients while the MGSr forecasts monthly mean attendances across CCG populations of tens to hundreds of thousands. Aggregation across both time and space suppresses variance, and the MGSr’s 3% MAPE is consequently not evidence of a superior model relative to Vollmer et al.’s [25] 6.8–8.6%. Aggregation across time and space smooths the fluctuations that make individual forecasts hard, so the difference in reported MAPE reflects how much the underlying data was smoothed before modelling began, rather than how well either model performed. This benchmarking error recurs across otherwise careful studies suggesting it is a weakness in how this literature reports comparative performance.

During cross-validation, models are trained only on observations preceding the test window and training set sizes are fixed through a rolling origin, which eliminates the issue whereby an expanding window might favour algorithms that benefit disproportionately from additional data. This design sets the methodology for the temporal validation used in the lagged-feature models of the present study.

The paper’s untested assumption is that it explicitly tests whether an equivalence between linear and tree-based methods remains even after the introduction of demographic or population-health variables. Every covariate in its model matrix is autoregressive, calendar-based or environmental, and none captures cross-sectional variation in population characteristics. Thus, the conclusion is only specific to its particular features, and testing its boundary is a substantial contribution.

2.4 Poursoltan et al. (2025)

Poursoltan et al. [17] forecast ED boarding volumes, where the number of admitted patients who remain in the emergency department while waiting for an inpatient bed, across two University of California San Diego Health emergency departments, differs from the target of attendance volume. Boarding reflects downstream pressures such as discharge rates, surgical throughput, and inpatient bed availability, whereas attendance measures demand for emergency care. The two are therefore different quantities with different determinants, so their error metrics should not be compared in the same table.

Six approaches are assessed against a two-week moving average model already deployed as a baseline, which consists of vector autoregression, XGBoost, the TimesFM model developed by Google, as well as hybrid models such as VAR+XGBoost and TimesFM+XReg. All the approaches are trained every day on a rolling window and predict for one to four days ahead. The input data consists only of operational and short-term features which contains historical boarding counts, scheduled surgeries, hospital census, and weekday indicator, without any demographic or population health level whatsoever. Therefore, this research occupies one end of the granularity scale with respect to the MGSr, addressing whether the separation by granularity is a good architectural feature or just a solution for this type of data.

The headline is that VAR+XGBoost has the lowest RMSE in all scenarios, lowering error rates by 16–42% from baseline at La Jolla and 5–19% at Hillcrest. The more important finding is that VAR without machine learning alone is only 0–3% higher than the combination of the two approaches in both locations. Thus, the contribution of the machine learning element is very minor, and it is also directionally inconsistent as it outperforms baseline at La Jolla, yet it underperforms baseline at Hillcrest. According to the authors, this discrepancy results from overfitting to covariates, although they have not tested it empirically. Even in the absence of any training at the new location, TimesFM still underperforms the two-week moving average at La Jolla; adding covariates using the XReg extension improves performance at one location, but not at another. In total, there is a consistent pattern, the classic econometric model does most of the predictive work, while the machine learning and foundation models add complexity without proportionate gains, across all six models.

With regard to the MGSr, the justification for two-level stacking is that the added complexity should yield proportionate accuracy gains, and Poursoltan et al. [17]’s results show that when tested against an

optimised simple baseline, such gains often fail to materialise.

As the study was conducted prospectively, another shortcoming relates to the independence of observations, which the authors themselves could not address due to the nature of their study. Forecasts generated via the VAR model were sent via email to the health system’s Mission Control unit every morning and used as the basis for accelerating patient discharges, transferring patients within the system and rearranging surgery appointments. This means the model not only predicted the boarding volume, but it was also affected, as the boarding volumes used to evaluate the model are dependent on the model’s forecasts, and the model’s forecasts cannot be separated from the interventions they enabled when computing the error. The authors claim the model’s calibration was stable, as 80% of actual values fell within the prediction intervals.

This inference is weak, as stable calibration could simply mean that the operational interventions were small relative to the underlying variation in the data. Distinguishing between these explanations would require a period in which forecasts were generated without being deployed, which the study does not provide. The MGSR does not face the same issue because its data entirely predate the modelling exercise, and this contrast matters when considering what different validation designs can and cannot establish.

A further limitation is the four-month validation window from July to October 2024, as the monthly RMSE varied by more than 20% at La Jolla and more than 50% at Hillcrest during this period. As the model was evaluated only through summer and early autumn, the validation does not capture winter surges, influenza season, or holiday-period dynamics. The variation already present within the four-month window also suggests that seasonal generalisability should not be assumed.

2.5 Synthesis

Across these four studies and the MGSR, four patterns are found, and together they frame the review and identify the gap.

The accuracy reported is more resolution-dependent than model quality, as monthly CCG-level means (MGSR, MAPE 3%) appear to be more accurate than daily hospital-level attendances. (Vollmer et al. [25], 6.8–8.6%), daily arrivals (Sudarshan et al. [23], 8.04–9.24%), hourly occupancy (Cheng et al. [5]) or daily boarding volumes (Poursoltan et al. [17]), but the ordering follows the degree of spatial and temporal aggregation almost exactly. Aggregation reduces variance, and comparisons that fail to account for this systematically flatter the coarsest models; this reporting issue has been noted in earlier systematic reviews of ED forecasting [26, 22, 12], and correcting it is a prerequisite for any meaningful cross-study benchmarking.

Model complexity is not always better than a well-specified baseline. Poursoltan et al. [17] find VAR within 0-3% of a VAR+XGBoost hybrid, Vollmer et al. [25] find penalised linear regression matching or beating tree ensembles across every horizon, Sudarshan et al. [23] report deep learning margins under two percentage points without significance testing. The exception is informative: when the predictor set is heterogeneous and cross-sectional rather than autoregressive, as in the MGSR’s Population Health model, tree ensembles do deliver significant gains. The value of complexity is thus dependent on feature structure, and claims of architectural improvement require an optimised simple baseline to be credible.

Predictor sets are operational and structural, and there are few studies covering both. Poursoltan et al. [17] use operational variables only, lagged occupancy with calendar regressors was used by Cheng et al. [5] and weather and search-query data by Vollmer et al. [25], but these are within a daily, autoregressive regime. Sudarshan et al. [23] add meteorological and calendar features to daily arrivals. James et al. [11] are the only ones who consider population health that varies annually along with monthly service capacity. This is the architectural contribution that MGSR makes. Fritsch et al. [8] and Rostami-Tabar and Ziel [20] identify a third type of demand driver. An event-driven demand shocks from sporting, cultural and public events, which is not systematically incorporated in any of the forecasting studies reviewed here.

Validation design is more diverse than the metrics reported and the rolling-origin temporal cross-validation of Vollmer et al. [25] is the most rigorous and transferable. Poursoltan et al.’s [17] prospective deployment is operationally realistic, but introduces a feedback confound that cannot be resolved within their design. One year at one site is validated by Cheng et al. [5]; Sudarshan et al. [23] do not report significance testing across configurations. Repeated five-fold cross-validation as used by James et al. [11] is suitable for assessing cross-sectional variance explained but does not respect temporal ordering and therefore cannot serve to establish forecasting performance as measured by the other studies.

Such patterns point to a clear gap in the literature. No existing studies examine whether the multi-granular architecture survives the structural break caused by the pandemic, whether it performs at a

finer geographic scale, whether Random Forest is the most suitable ensemble method and whether the population health layer maintains its usefulness after the discontinuation of the ONS Health Index. All three of these questions are testable with publicly available data. Together they form the basis of this study.

Chapter 3

Replication of the MGSR Model

This chapter describes the replication of the multi-granular stacked regression (MGSR) model proposed by James et al. [11]. The aim was to reconstruct both the dataset and the model from publicly available sources, verify that the reported results can be reproduced, while establishing a validated baseline from which the extension work in Section 4 builds. The replication involved two distinct stages: first, assembling a CCG-level panel dataset from raw NHS and ONS publications; and second, implementing the MGSR architecture and evaluating it using the same cross-validation methodology described in the original paper.

3.1 Dataset Construction

3.1.1 Overview

James, Wood and Denholm (2023) [11] use a carefully selected dataset of monthly observations for 74 Clinical Commissioning Groups (CCGs) in England between January 2018 and December 2019. Each observation contains eight independent variables representing capacity of urgent care services (GP appointments available, NHS 111 calls offered, and ambulance calls answered) and population health characteristics (total population, the percentage of the population aged 65 and over, and the ONS Health Index domains People, Places, and Lives and a single dependent variable, monthly ED attendances. All count-based variables are expressed as rates per 10,000 population to allow comparability across CCGs of different size.

The authors provide their processed dataset ("master_scaled.csv") on GitHub, but do not publish the code they used to build it from raw sources. Therefore, one requirement was the independent replication of the dataset to check the integrity of the original data as well as to set up a pipeline that could be extended later to include post-2019 data. Table 3.1 provides a summary of the sources of raw data used in the replication.

3.1.2 GP Appointments

The data on GP appointments was the most straightforward variable to recreate, as NHS Digital publishes monthly Appointments in General Practice files at CCG level, so no geographic mapping was needed. The total number of appointments per month was taken from the CCG level summary sheet and scaled by the CCG population (in units of 10,000) to produce the scaled variable `gp_appt_available`.

A complication faced was that this publication series started only in October 2018 so, there were no values for January to September 2018 from the raw source. During the final merge step, the nine-month period was filled with corresponding values of the authors published dataset. The 73 CCGs included in the analysis were treated independently, starting from October 2018.

3.1.3 Ambulance Data

Ambulance data is published in the AmbSYS Time Series which reports monthly call volumes for each of the 11 ambulance trusts covering England. The disaggregation of the trust-level figures was required as each trust serves multiple CCGs, and this was achieved by employing population-share stratification: if a CCG represents 30% of a trust's total covered population, it receives 30% of that trust's monthly call volume.

Table 3.1: Data sources used in the dataset replication.

Variable	Source	Geographic level published	Processed independently	Using authors' values
GP appointments available	NHS Digital — Appointments in General Practice	CCG	73	0
Ambulance calls made/answered	NHS England — AmbSYS Time Series	Ambulance trust (11 trusts)	64	9
NHS 111 calls offered/answered	NHS England — NHS 111 Minimum Dataset	111 area (53 areas)	48	25
Population	ONS Mid-Year Population Estimates (SAPE21, SAPE22)	CCG (via ONS codes)	64	9
% aged 65 and over	ONS Mid-Year Population Estimates (SAPE21, SAPE22)	CCG (via ONS codes)	64	9
People, Places, Lives	ONS Health Index (via LSOA crosswalk)	Local authority	73	0

The CCG to ambulance trust mapping was extracted from a lookup sheet contained within the AmbSYS file itself, after which standardising trust identifiers identified 64 of the 73 CCGs that could be mapped to Sub-ICB.

The other 9 CCGs used non-standard NHS codes that were not contained within any publicly available lookup table and the data was obtained from the authors' dataset.

The two extracted variables were `amb_sys_made` (total calls made) and `amb_sys_answered` (total calls answered, category A1), scaled per 10,000 population. The paper only used `amb_sys_answered` as a model input, but both were kept in the replicated dataset for completeness.

3.1.4 NHS 111 Data

NHS 111 call data is published in the NHS 111 Minimum Dataset at the level of 111 service areas (53 areas across England) and not at CCG level. For ambulance data the same population-share stratification approach was followed: a mapping from CCGs to 111 areas was derived from a lookup sheet within the MDS file and call volumes for each area were apportioned to CCGs in line with their share of the area's population.

This variable had the highest fallback rate of all variables in the dataset. For 48 of the 73 CCGs, it was possible to map to an area using the published mapping. The remaining 25 were not present in the lookup, likely because they were supplied by providers not included in the mapping or because of a change in CCG boundaries since the mapping was produced. The authors' published values were used for these 25 CCGs.

The two variables extracted were `111_111_offered` (calls offered to the service), and `111_111_answered` (calls answered), both scaled per 10,000 population. Like with the ambulance data, the model uses '111_111_offered' as input.

3.1.5 Population

Population estimates are from the ONS Mid-Year Population Estimates for 2018 (SAPE21) and 2019 (SAPE22). These files contain population by CCG using ONS geographic codes (E38 series), which had to be mapped to the NHS CCG codes (three-character alphanumeric codes such as 00Q) used elsewhere in the dataset. The mapping was taken from the CCG lookup sheet in the AmbSYS file which has both code systems listed side by side.

64 CCGs are linkable following this mapping. The other 9 used non-standard codes not found in the lookup, with population values taken from the authors' file. A small number of CCGs appeared in only one year as a consequence of boundary changes – these were forward or backward filled so that every CCG had a population value for both 2018 and 2019. The final population variable is in 10,000s.

3.1.6 Health Index Crosswalk

The ONS Health Index, which generates annual scores for the three dimensions of People, Places, and Lives, is available at the Local Authority (LA) level. But for the MGSR model, this score needs to be generated at the Clinical Commissioning Group (CCG) level. This necessitated a crosswalk because LAs and CCGs are two separate geographic groupings and do not nest nicely into each other.

For the creation of the crosswalk, we used the ONS LSOA to CCG to LA crosswalk table. Every Lower Super Output Area (LSOA) in England falls under one LA and one CCG. By calculating how many LSOAs are common between any LA-CCG combination, we were able to generate proportional weights. If an LA consists of 100 LSOAs and out of those 60 LSOAs belong to a particular CCG, the weight for that particular CCG will be 0.60.

This approach produced CCG-level Health Index scores for all 73 CCGs in both 2018 and 2019. To verify the crosswalk, the resulting values were compared against the authors' published scores. The correlation between the replicated and original values exceeded 0.98 for all three domains (People, Places, Lives), with small mean absolute differences reflecting minor variations in the exact LA-to-CCG mapping vintage used.

3.1.7 ED Attendances

The data regarding the number of ED attendance is provided by NHS England, on a trust level, not a CCG level. The mapping of ED attendance numbers from trust level to CCG level needs trust to CCG concordance, which is not provided by the authors. The process of such a mapping is non-trivial since some trusts have a patient population served across several CCG borders. Thus, due to complexity and a possibility of introducing errors during mapping, the ED attendance values were used as provided in the published dataset of the authors.

3.1.8 Final Merge and Validation

All variables independently processed were combined using the CCG code, calendar month and year with the authors' ED attendance values serving as the starting point. GP values which had gaps during January-September 2018 were imputed based on the author's file. After deleting all rows having any missing values, the total number of observations in the final replicated file was 1,435, which is close to the 1,618 observations in 74 CCGs presented in the original study. The slightly different CCG counts and row counts can be attributed to the slight variations in how missing values were handled.

For verification purposes, the replicated values for one particular CCG (code 00Q) were compared with the authors' published values. The values for GP appointment and ambulance services matched perfectly for the independently processed CCG.

3.2 Model Implementation

3.2.1 Architecture

The MGSR model was developed based on the architecture provided in James, Wood, and Denholm (2023), namely, their Figure 1. It includes three models stacked in an ensemble:

Level 0a - Capacity Model. A Random Forest regressor for predicting mean monthly ED attendances using four of the five annually varying variables (i.e., population, People, Places, Lives), omitting the percentage aged 65 and over that the original paper includes. The parameters were set to `max_depth=5` and `n_estimators=6`, which corresponded to the settings provided by the authors as being selected in order to maximise validation performance and minimise model complexity.

Level 0b - Population Health Model. A Random Forest regressor for predicting mean monthly ED attendances using four variables that vary annually (i.e., population, People, Places, Lives). As annual population health variables and monthly ED attendances are used here, it was necessary to aggregate data into one observation per CCG and year, averaging monthly ED attendances for this observation. As a result, the sample size was reduced from approximately 1,400 monthly observations to approximately 128.

CCG-year groups. The hyperparameters were set to `max_depth=4` and `n_estimators=2`, again following the original paper.

Level 1 - Combined Model. An Ordinary Least Squares (OLS) linear regression that takes the predictions of the Capacity Model and the Population Health Model as its two inputs and produces a final prediction for monthly ED attendances. OLS was chosen due to the small number of inputs (two) and the expected linear relationship between the Level 0 predictions and the true values.

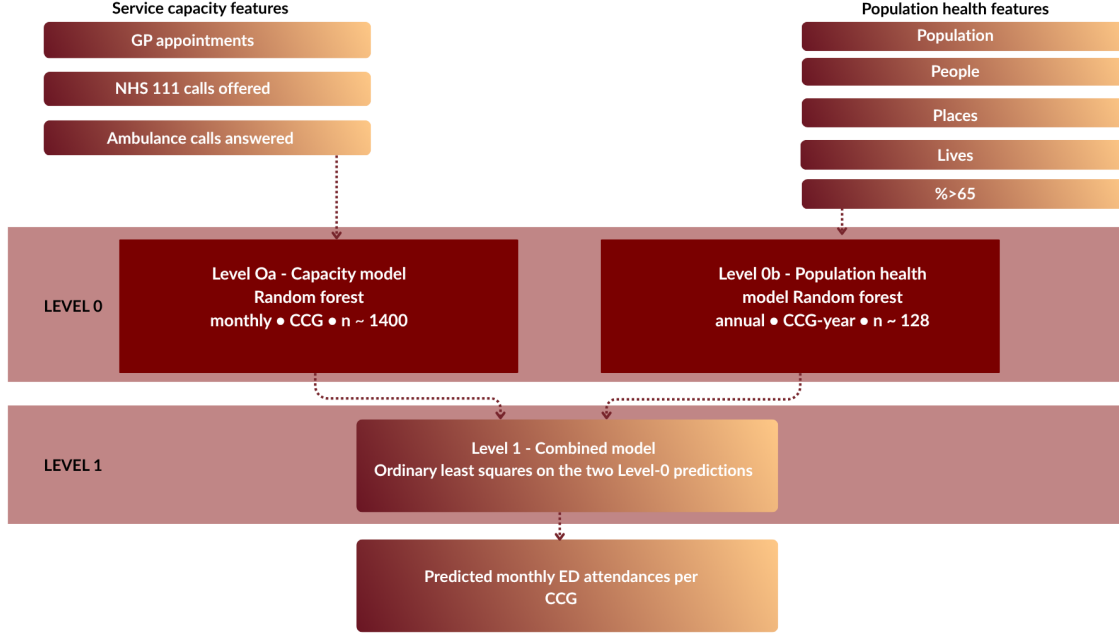


Figure 3.1: The multi-granular stacked regression architecture, reproduced from the specification of James et al. [11]. Two Level-0 random forests are estimated on feature blocks that differ in both temporal and geographic granularity; their out-of-fold predictions are combined by an ordinary least squares model at Level 1.

All models were implemented in Python using scikit-learn, consistent with the original study.

3.2.2 Cross-Validation Protocol

Performance of the models was evaluated by applying 5-fold cross-validation for five iterations, resulting in 25 validation folds in total. For each fold, the data set was split into a training set (80%) and a validation set (20%), where every data point was used once for validation in every iteration. An average R-squared statistic was calculated based on 25 validation sets.

For the Combined MGSR model, one extra measure was taken to avoid possible data leakage from one level of the model to another. Namely, the Population Health Model was estimated separately for every training fold using an inner 5-fold cross-validation loop so that predictions of the training data based on it were actual out-of-fold predictions and not in-sample predictions. It is in line with the method described in the original paper, where the Population Health model was estimated separately for the Combined model.

3.2.3 A Note on the Population Health Model Grouping

Another methodological consideration that was particularly important was the grouping function used for the Population Health Model. As can be seen, the initial codebase uses `population` as the grouping variable, which means that each group is defined according to its unique value of population. In reality, however, due to the fact that each CCG has its own value of population for each year considered, this results in the creation of around 128 groups out of 73 CCGs and a partial two-year period, where some of them have identical population values.

When replicating the model, we decided to use a more explicit grouping approach based on both CCG code and year, thus ensuring that there would be exactly one group per each CCG-year combination irrespective of whether the population values coincided or not. Such an approach resulted in a similar size and structure of grouped data. Nonetheless, it is quite likely that the discrepancy in grouping approach, along with some other differences between CCGs (as they are either shared or removed differently),

contributed to the difference in R^2 value. Most importantly, the replication omitted the percentage of the population aged 65 and over from the Population Health Model, a variable the original paper includes as one of its five population health inputs. Given that age structure is a strong predictor of emergency department demand, this omission is likely the primary contributor to the lower R^2 observed here, ahead of the data and grouping differences noted above

3.3 Results

3.3.1 Individual Model Performance

Table 3.2 summarises the cross-validated R^2 for each component of the MGSR, comparing the replicated results against the values reported in the original paper.

Table 3.2: Cross-validated R^2 for each component of the MGSR.

Model	Paper R^2	Replicated R^2	Delta
Capacity (Level 0a)	0.41 ± 0.04	0.42 ± 0.04	+0.01
Population Health (Level 0b)	0.62 ± 0.22	0.54 ± 0.30	-0.08
Combined MGSR (Level 1)	0.79 ± 0.02	0.76 ± 0.03	-0.03

The Capacity Model had a similar performance with a mean test R^2 of 0.42 compared to the reported 0.41. The ordering of feature importance differed from the original: the replication found 111 capacity to be the most important variable (mean Gini importance of 0.52), followed by ambulance capacity (0.28) and GP capacity (0.19), whereas the original paper reported ambulance capacity as the most important (GI 0.65), with GP capacity least important (GI 0.16). The mean R^2 for the Population Health Model was 0.54 which was somewhat lower than the reported 0.62, and the standard deviation was higher (0.30 vs. 0.22). The high variance in both cases suggests that the approach’s performance is very sensitive to the specific CCGs included in the training and validation sets, which is unsurprising given the small effective sample size of approximately 128 pooled observations. In order to prevent overfitting, the model is kept small by setting `n_estimators=2` and `max_depth=4`, however, it also makes it prone to changes in the makeup of each fold. In this regard, we would like to mention that this problem is highlighted by the authors themselves, who used only 145 observations to train the Population Health Model, leading to inaccurate predictions of the CCGs.

Several reasons are likely for the difference between the Population Health R^2 s reported and replicated. First, the replicated dataset has 73 CCGs vs. 74 in the original and 1,435 rows vs. 1,618 rows, indicating slightly different treatment of missing data. Second, the implementation of the grouping function is subtly different, as described above.

Third, while the values of the Health Index obtained using the LSOA crosswalk were highly correlated with the authors’ ($r > 0.98$), they are not identical and the small residual differences could be magnified in a model trained on fewer than 150 data points. The combined MGSR still produced a mean test R^2 of 0.76, close to the reported 0.79 and representing a substantial improvement over either component model alone. The stacking mechanism is successful at combining the complementary information associated with the two levels: the Capacity Model is able to capture within-CCG monthly variation driven by service demand patterns and the Population Health Model is able to capture between-CCG variation driven by demographic and health differences.

3.3.2 Predicted versus Actual

The predicted vs actual ED attendances (per 10,000 population) for all out-of-fold predictions were plotted using a scatter plot. We saw a strong positive relationship with points clustered around the 45-degree line and more scatter at the extremes of the distribution as expected. Consistent with the discussion in the original paper, the model under-predicts CCGs with very high ED attendance rates and over-predicts very low ones. This is a well-known problem with the Random Forest regressor, which cannot extrapolate outside its training data range.

3.3.3 Discussion of Replication Fidelity

The replication was successful in reproducing the overall structure and qualitative findings of the MGSr. The Capacity Model was very similar to the original in performance and feature importance rankings. The Population Health Model and Combined MGSr fell slightly short of the reported R^2 values but tracked them closely. The sign and relative size of effects were consistent, with population and the Lives domain being the most important population health variables, and the stacked combination outperformed either level alone.

These differences are probably caused by the small changes in the data (fewer CCGs, different treatment of missing values, and approximate crosswalk used in the Health Index) as well as the instability of the model based on less than 150 grouped observations. All these differences should not be seen as any methodological problems but rather as the sensitivity of the Population Health Model to even very slight changes in input data, which is also admitted by the original authors as one of the limitations that can be overcome by using more data.

Having developed the baseline for the replicated Population Health Model, the next chapter develops the MGSr approach in two directions. First, it uses the additional variable %>65 as described in the paper, which was omitted in the first replication. Secondly, it uses XGBoost and Gradient Boosting as alternatives to Random Forest at Level 0. On the same CCG data, substituting Gradient Boosting for Random Forest at Level 0b raises the Population Health R^2 from 0.54 to 0.70, motivating the tree-ensemble comparison carried into Chapter 4.

Chapter 4

Extension: Applying the MGSR to Post-2022 Sub-ICB Data

4.1 Data Challenges

The extension of the MGSR into the post-COVID period (April 2022 - December 2024) was faced with three structural discontinuities which collectively did not allow an easy rerun of the original pipeline on recent data.

First, the replacement of CCGs with 42 Integrated Care Boards in July 2022 reduced the number of commissioning units from 106, following successive mergers of the original 200-odd CCGs, to 42. The admissions dataset that has been published aggregates attendances only to ICB (System) level and does not contain a published breakdown by Sub-ICB. An initial attempt to recover Sub-ICB resolution by proportionally disaggregating ICB-level attendances according to population was methodologically unsound because the population-based splitting and subsequent per-capita scaling cancelled each other algebraically. That data set does not contain a published breakdown by Sub-ICB. An initial attempt to recover Sub-ICB resolution by proportionally disaggregating ICB level attendances according to population was methodologically unsound: the population-based splitting and subsequent per-capita scaling cancelled each other algebraically. To see why, let Y_j be the published count for parent geography j , P_i the population of Sub-ICB i within it, and $P_j = \sum_i P_i$ the parent total. Population-share disaggregation assigns $\hat{Y}_i = Y_j P_i / P_j$ to each Sub-ICB. Scaling to a per-capita rate gives

$$r_i = \frac{\hat{Y}_i}{P_i/10,000} = \frac{Y_j \frac{P_i}{P_j}}{\frac{P_i}{10,000}} = \frac{Y_j}{P_j/10,000} = r_j, \quad (4.1)$$

so P_i cancels and every Sub-ICB within the same parent receives an identical rate r_j . Diagnostic analysis confirmed this: the within-ICB standard deviation of the resulting per-capita target was 0.21 on a mean of about 440, virtually zero variance, and cross-validated R^2 was consistently negative irrespective of model family, since the features were different across Sub-ICBs while the target was not.

Secondly, the ONS Health Index was discontinued after the 2021 release. The People, Places and Lives domain scores that made up the population health feature set in the original paper have no direct successor publication for 2022 onwards. The extension therefore needed an alternative source of population health indicators.

Third, the original capacity variables lost some of their Sub-ICB resolution. NHS 111 call volumes are published at contract level (around 40 contracts, broadly aligned to ICBs) and ambulance activity at the level of 11 regional services. Both are disaggregated to Sub-ICBs by the same population-share stratification and therefore both have the same cancellation when scaled per capita. Only GP appointment data published natively at Sub-ICB level by NHS Digital has any real within-ICB variance.

4.2 Data Sources

We used the Hospital Episode Statistics Monthly Activity Report, an open data CSV published monthly by NHS Digital, to resolve the target variable. HES data are derived from the Emergency Care Data Set with each attendance coded with the patient's home Sub-ICB code. The HES open data have "Organisation

breakdown = CCG" rows with monthly emergency admission counts per Sub-ICB based on patient-level records, not population splitting. This yielded 97 Sub-ICBs with complete target data from April 2022 to December 2024. After merging with population data, this resulted in 3,298 monthly observations, reduced to 3,201 once duplicated records were removed (Section 4.3). The per-capita target varies genuinely within ICBs: its mean within-ICB standard deviation is 23.60 against an overall standard deviation of 26.52, a ratio of 0.89, which confirms real variation at Sub-ICB level and validates the data source.

Population health indicators from the Office for Health Improvement and Disparities Fingertips platform were used instead of the Health Index using the `fingertips_py` API client [18]. Seven QOF chronic disease prevalence indicators were chosen: asthma, COPD, coronary heart disease, depression, diabetes, heart failure and hypertension. The data for QOF is taken from disease registers held by primary care GPs and is therefore independent of the emergency admissions target which is derived from patient records held in secondary care [13]. These indicators are published by Fingertips at ICB level, and were mapped to Sub-ICBs via the parent ICB, so that Sub-ICBs within the same ICB have the same health feature values. Post-merging coverage was 77% of observations; the remaining were lost to ICBs that could not be mapped using the ONS-to-NHS code bridge.

The Sub-ICB level GP appointment data were taken from Appointments in General Practice and after merging we achieved 94% coverage. Population and age structure were derived from ONS mid-year Sub-ICB population estimates for 2022, 2023 and 2024. The percentage of the population aged 65 and over was calculated as in the original paper.

4.3 Model Implementation

Before modelling, the constructed panel was tested for duplicated observations; ninety-seven rows, one per Sub-ICB, were found to be duplicates of the November 2022 record. The duplicated rows were byte-identical, arising from overlapping records in consecutive HES annual files rather than any genuine source disagreement. Removing them reduced the panel to 3,201 observations from 3,298 observations, which resulted in a balanced panel of 97 Sub-ICBs observed over 33 consecutive months from April 2022 to December 2024. Had we kept them, the same observation would have been in both the training and validation portions of a cross-validation fold.

Further comparisons were conducted on a single complete-case sample to provide a consistent basis for evaluating model performance. Since the Fingertips QOF indicators are merged for only 77.3% of rows and GP appointments for 93.9%, restricting each model to the rows on which its own features are observed would evaluate different specifications on different subsets of the data, and make their R^2 values non-comparable. The analysis sample is thus set to 2,325 observations for 75 Sub-ICBs in 25 ICBs. All the models, baseline and significance tests below are assessed on these same rows with an identical set of cross-validation folds.

The architecture of the MGSR was re-built as per James et al. (2023) [11], with the Level-0a submodel occupying the same structural position as the published capacity model but using a different Sub-ICB feature set, so two definitions are evaluated in parallel. The first, described below as the capacity block as specified, combines GP appointments per 10,000 population with seasonal encoding; it is the closest available analogue of the published block since NHS 111 and ambulance activity are published above Sub-ICB level and lose all within-ICB variance under population-share disaggregation.

The second, referred to as the extended service activity block, incorporates four additional activity measures for secondary care, which come from the same HES Monthly Activity Report as the target: elective admissions, first outpatient attendances, outpatient DNAs, and GP referrals, all per 10,000 population. These are termed as service activity and not capacity because, in the case of elective admissions and outpatient attendances, they are co-produced in the same hospital system as the target and, therefore, do not indicate the diversionary capacity of upstream urgent care that the initial block sought to measure. Both definitions must be reported since, as shown in Section 4.4, conclusions regarding the stacked architecture depend on which is chosen.

The Level-0b submodel is a tree ensemble trained on the seven Fingertips QOF prevalence indicators, along with population and proportion of the population over age 65. Level 1 is an ordinary least squares regression based on the predictions made by the two Level-0 models. All tree-ensemble hyperparameters are tuned once per model role using `RandomizedSearchCV` [2] with 30 configurations, scored on the same repeated 5-fold cross-validation splits used to report every downstream result. Five specifications are tuned separately, because the full model, the Level-0a service-activity block and the Level-0b health-and-demographics block operate on different, differently-sized feature sets:

(i) the full 16-feature model for RF, GBR and XGBoost, used across Sections 4.4 and 4.5; (ii) RF and GBR on the Service Activity block for the Level-0a role in the MGSr stack

Table 4.1: Tuned hyperparameters by model role. All models use `random_state=0`.

Role	Learner	n_estimators	max_depth	learning_rate	subsample
Full model	GBR	476	5	0.096	0.96
Full model	XGBoost	484	5	0.050	0.73
Level-0a	GBR	287	5	0.033	0.65
Level-0b	GBR	344	4	0.154	0.79

Role	Learner	n_estimators	max_depth	min_samples_leaf	max_features
Full model	RF	265	9	2	0.64

(iii) RF and GBR on the Health + Demographics block for the Level-0b role. Table 4.1 reports the chosen configurations. Hyperparameters are selected once on the same folds later used for reporting rather than through a fully nested outer/inner cross-validation, which is a standard simplification at this sample size but means the reported R^2 is mildly optimistic relative to a strictly nested design for reproducibility.¹

There is one implementation detail that differs from the previous version of this analysis, but is in closer accordance with the original paper. The Level-1 meta-learner needs to be trained on Level-0 predictions which were made on instances for which the Level-0 models do not know the target values; training the Level-1 model on in-sample Level-0 predictions would result in biased estimates of the stack’s performance. Thus, the Level-0 predictions for training the Level-1 meta-learner are made inside an inner five-fold cross-validation loop within each outer training fold. Both versions are computed so that the size of any leakage is visible. At this sample size it is negligible: the out-of-fold and in-sample meta-estimates differ by less than one fold-standard-deviation in every configuration (capacity block, gradient boosting: out-of-fold $R^2 = 0.859$ against 0.826 in-sample; extended block: 0.875 against 0.846), and the sign of the difference is not stable across configurations. This is expected; the shallow Level-0 learners and the two-input ordinary least squares meta-model have little capacity to memorise training targets, so fitting the meta-learner in-sample offers little to leak. The out-of-fold predictions are used throughout as the conservative choice, consistent with the original paper’s procedure, but at this sample size the distinction has no material effect on the reported figures.

The models were trained using five-fold cross-validation with three iterations, yielding a total of fifteen validation folds. The fold numbers were determined once and used uniformly across all specifications to allow a paired comparison of fold-level R^2 and, hence, a Wilcoxon signed rank test of model performance differences. Two baselines are reported in addition to every model specification: the global-mean baseline, which predicts the mean value of the training set for all observations, and the group-mean baseline, which uses the training-set mean of each Sub-ICB as its prediction. The latter is the more interesting baseline here because a model failing to outperform it would only have succeeded in recovering the identity of each area and nothing else. The robustness to geography was investigated using GroupKFold cross-validation, with the Sub-ICB’s and even the ICB’s held-out entirely from the training set. The forecast quality was investigated by training a model using April 2022 – December 2023 data and forecasting the twelve months of 2024, using activity features lagged one month ahead.

4.4 Results

Since the rate of elective admissions is such an important ratio in this study, it deserves to be carefully examined before its use in the regression. The reason behind correlation between the two ratios sharing the same denominator is partially built-in. Thus, three additional tests should be done. Firstly, correlation between unscaled counts is 0.96; however, this fact is determined by the fact that all four activity counts are correlated with the target count between 0.81 and 0.96. Secondly, partial correlation between the two rates, taking into account the control for population, is 0.48 and is almost equal to the unadjusted 0.53. Thirdly, and, most importantly, the correlation between two rates within the Sub-ICB, when the two series are de-meant by the area, i.e., the cross-sectional component is removed, is 0.50. Months when

¹Full model specifications are given in the accompanying analysis notebook.

Table 4.2: Pearson correlation of each feature with the emergency admission rate per 10,000 population. Within-area correlations are computed after de-meaning both variables by Sub-ICB, and are shown only for features that vary within areas (activity and GP features); annual QOF and structural variables are constant within a Sub-ICB-year and have no meaningful within-area correlation.

Feature	Cross-sectional r	Within-area r
Elective admissions rate	0.53	0.50
COPD prevalence	0.35	—
Hypertension prevalence	0.32	—
Heart failure prevalence	0.31	—
CHD prevalence	0.29	—
GP appointment rate	0.24	0.28
Outpatient DNA rate	0.22	0.30
Diabetes prevalence	0.22	—
Asthma prevalence	0.20	—
Depression prevalence	0.19	—
Pct. aged 65+	0.13	—
Outpatient first attendance rate	0.07	0.35
GP referrals rate	0.05	0.28
NHS 111 offers rate	−0.01	—
Population (10,000s)	−0.28	—

one Sub-ICB has more elective activity are the same months with more emergency activity, which cannot be done by the common denominator.

Tests reveal relationship which is not obvious from the raw correlations. First, outpatient first attendance and GP referral are very loosely correlated with the target in areas ($r = 0.07$ and $r = 0.05$), but have a much higher correlation within areas ($r = 0.35$ and $r = 0.28$). The former is hidden by the heterogeneity in area referral practices, but the latter relationship really exists. This discrepancy between correlation in areas and within areas is not accounted for the original paper and becomes important in prediction rather than explanation. There is one variable which affects both the target and control measure: the number of working days per month affects both elective and emergency care simultaneously. This variable can be introduced as a covariate.

Feature set comparison. Eight model families were assessed across eight feature sets on the common analysis sample; Table 4.3 reports the full grid. Using health features alone provides $R^2 = 0.32$. Using GP capacity alone is uninformative on its own (negative under every tree ensemble). Using population and percentage aged 65 and over alone boosts this up to 0.86, and adding health features and seasonal encoding raises it to 0.88. The extended service-activity block alone provides 0.55, and the full set provides 0.91.

Table 4.3: Cross-validated R^2 by model family and feature set (repeated 5-fold, 3 repeats, 15 folds; $n = 2,325$ observations, 75 Sub-ICBs). Tree-ensemble hyperparameters are tuned once per role (see Section 4.3). Baselines are computed on the same folds.

Model	Health only	GP only	GP + Health	Demog. only	GP + H + D	GP + H + D + S	Service activity	All features
Ridge	0.16	0.04	0.20	0.07	0.22	0.22	0.41	0.45
Lasso	0.16	0.04	0.20	0.07	0.22	0.22	0.41	0.45
ElasticNet	0.16	0.04	0.20	0.07	0.22	0.22	0.40	0.44
KNN ($k = 10$)	0.07	−0.07	−0.05	0.84	0.05	0.05	0.18	0.22
SVR (RBF)	0.29	0.03	0.40	0.17	0.51	0.49	0.49	0.74
Random Forest	0.32	−0.03	0.43	0.80	0.84	0.85	0.56	0.85
Gradient Boosting	0.32	−0.33	0.27	0.86	0.82	0.88	0.55	0.91
XGBoost	0.32	−0.08	0.36	0.85	0.85	0.89	0.57	0.92

Baselines: global mean −0.00; group mean (each Sub-ICB’s own) 0.81 ± 0.03 ; Sub-ICB-mean ceiling 0.82.

The apparent contribution of the demographic features is the core finding of this section, and it does not reflect an actual predictive relationship. Using population and percentage aged 65 and over and no

other features leads to $R^2 = 0.86$. A baseline that uses each Sub-ICB’s training-set mean, making no use of any features whatsoever, leads to $R^2 = 0.81 \pm 0.03$. And a predictor that perfectly knows the mean of every Sub-ICB, and so reaches the maximum possible performance for any purely area-level model, leads to 0.82. The demographic-only model has therefore exceeded the group-mean baseline and effectively matched the Sub-ICB-mean ceiling using only two features.

The reason is that population and age distribution serve as identifiers of area rather than covariates. Each Sub-ICB has a distinct population, and a diagnostic test shows that 100% of the unique static feature profiles of the sample map to a single Sub-ICB. The additional evidence is the performance of k -nearest neighbour regression, which attains $R^2 = 0.84$ on the demographics-only set, its maximum performance on any feature set in the study, while achieving scores ranging from -0.07 to 0.22 on all others. A k -nearest-neighbour regressor is a lookup table per construction; that it performs best exactly where the features identify the area is direct evidence that the quantity being recovered is the area mean. Under the untuned specification used in an earlier version of this analysis the demographic-only score was 0.82, matching the ceiling but not exceeding it; tuning raises it to 0.86, above the ceiling, because the tuned learner is now flexible enough to lightly interpolate around the area mean rather than only recover it.

This means that the headline figure should be reconsidered, as about 0.81 of the full model’s $R^2 = 0.91$ can be recovered using a lookup table of historical area means. The genuine additional contribution of the features is the 0.10 by which the model beats the group-mean baseline, and Wilcoxon tests confirm that this difference is not noise (gradient boosting, all features vs group-mean baseline: $\Delta R^2 = +0.101$, $p = 0.0001$; XGBoost: $\Delta R^2 = +0.105$, $p = 0.0001$). This difference comes from the activity features, which vary monthly and are thus the only features of the model capable of distinguishing months within an area. In contrast, the extended service-activity block alone significantly underperforms the group-mean baseline ($\Delta R^2 = -0.25$, $p = 0.0001$), because it does not possess the area-level information contained within the baseline.

The leave-one-feature-out analysis of the complete specification supports this interpretation, with removal of the proportion of the population aged 65 and over resulting in a highly significant loss of 0.029 in R^2 ($p < 0.001$). Removal of the elective admission rate loses 0.010 of R^2 , but the loss is not statistically significant ($p = 0.21$). Removal of population loses 0.007 of R^2 ($p < 0.001$); population is retained for consistency with the original specification, but it is the least well-justified variable in the model, since it is the denominator of the target and, because each Sub-ICB has a near-unique population, partly functions as an area identifier rather than as a measure of need. Excluding it entirely reduces the cross-validated R^2 from 0.911 to 0.904, so none of the results below depend on it. Removal of the seasonal encoding loses 0.008 of R^2 ($p < 0.001$); all other features lose less than 0.005 of R^2 , while removal of GP appointments, the sole feature retained from the capacity block of the original paper, does not materially change performance ($\Delta R^2 = -0.004$, $p = 0.015$). None of the seven QOF indicators has a measurable impact ($\Delta R^2 \leq 0.002$ each, all $p > 0.05$).

Table 4.4: Leave-one-feature-out analysis for the full gradient boosting specification ($R^2 = 0.911$). Each row shows the change in cross-validated R^2 when that feature is removed.

Feature removed	ΔR^2	p -value
Pct. aged 65+	-0.029	< 0.001
Month (seasonal encoding)	-0.008	< 0.001
Population	-0.007	< 0.001
Elective admissions rate	-0.010	0.21 (ns)
7 QOF indicators	≤ 0.002 each	all > 0.05 (none significant)
GP appt / outpatient / referrals	removal does not reduce R^2	—

Two additional tests support this interpretation. First, a permutation test, with the target randomly permuted, produces $R^2 = -0.22$ – showing that the fitted relationship is not spurious. Second, a temporal holdout, with training on 2022-2023 and prediction of 2024 with the same non-lagged specification, yields $R^2 = 0.56$ compared with the cross-validated 0.91. The difference of 0.35 between random and temporal validation indicates how much of the cross-validated score arises from the fact that the model has seen the area before.

Figure 4.1 shows the corresponding predicted-against-actual scatter for the temporal holdout.

Comparison of MGSR. Unlike the initial assessment, the stacked architecture outperforms its submodels in all tested configurations. In the case of the capacity block as defined, consisting of GP appointments

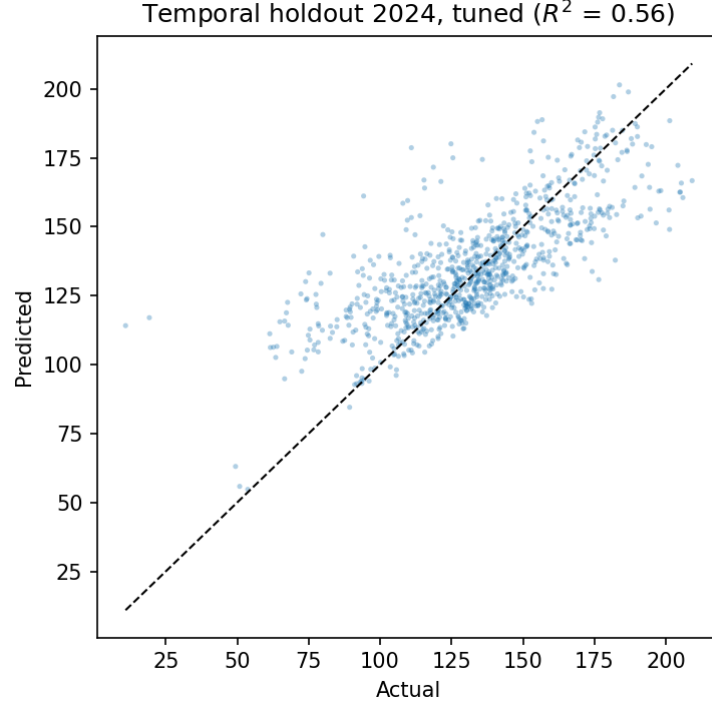


Figure 4.1: Predicted against actual emergency admission rates per 10,000 population, temporal holdout (trained on April 2022 – December 2023, evaluated on 2024). The dashed line is the identity.

and seasonality, the submodel itself is not predictive in its own right ($R^2 = -0.04$), the stack improves marginally on the Level-0b score, from 0.857 to 0.859. In the case of the extended service-activity block, where Level-0a attains a score of 0.57 by itself, the stack attains a score of 0.88, which is an improvement by 0.019 on the better of the two Level-0 models. With Random Forest Level-0 learners, the improvement is essentially the same: Level-0a score is 0.56, Level-0b is 0.84, and their combination is 0.86, an improvement of 0.020 on the better of the two Level-0 models. Figure 4.2 shows the pattern. This indicates that the complementary-information property extends to Sub-ICB level, at least where the capacity block carries signal, and thus the architecture is replicated on data the authors did not use.

Table 4.5: MGSR performance by Level-0a definition (repeated 5-fold, 3 repeats). β_{0a} and β_{0b} are the fitted Level-1 OLS weights. The final row reproduces the published figures for comparison.

Level-0a definition	Learners	L0a	L0b	Stack	β_{0a}	β_{0b}
Capacity block as specified	RF + RF	-0.03	0.84	0.85	0.18	1.05
Capacity block as specified	GBR + GBR	-0.04	0.86	0.86	0.13	0.98
Extended service activity	RF + RF	0.56	0.84	0.86	0.29	0.90
Extended service activity	GBR + GBR	0.57	0.86	0.88	0.27	0.84
<i>James et al. (2023), CCG 2018-19</i>	<i>RF + RF</i>	<i>0.41</i>	<i>0.62</i>	<i>0.79</i>	<i>0.22</i>	<i>0.85</i>

The coefficients estimated by the meta-learner provide a further point of correspondence. Under the extended service-activity block with gradient boosting Level-0 models, the ordinary least squares meta-learner assigns weight 0.27 to the capacity subprediction and 0.84 to the population health subprediction, close to the 0.22 and 0.85 reported in the paper. The capacity block as specified gives 0.13 and 0.98, the same ordering of preference, with more weight still on the population health signal. Table 4.5 reports all four configurations alongside the published figures. There is one test, however, that restricts the validity of the architecture. Against a single flat model trained on the union of the two feature sets, the stack performs significantly worse with gradient boosting (0.875 vs. 0.912; $p = 0.0001$) and is statistically indistinguishable from the flat model with Random Forest (0.863 vs. 0.852; $p = 0.073$). Under the untuned specification, included here as a robustness check against the tuned results, the flat model matched the stack (with GBR 0.870 vs. 0.859, $p = 0.12$) and was clearly worse with **Random** Forest (0.833 vs. 0.679, $p = 0.0001$), attributable to the shallow parameterization of Level-0b. With every learner now tuned to

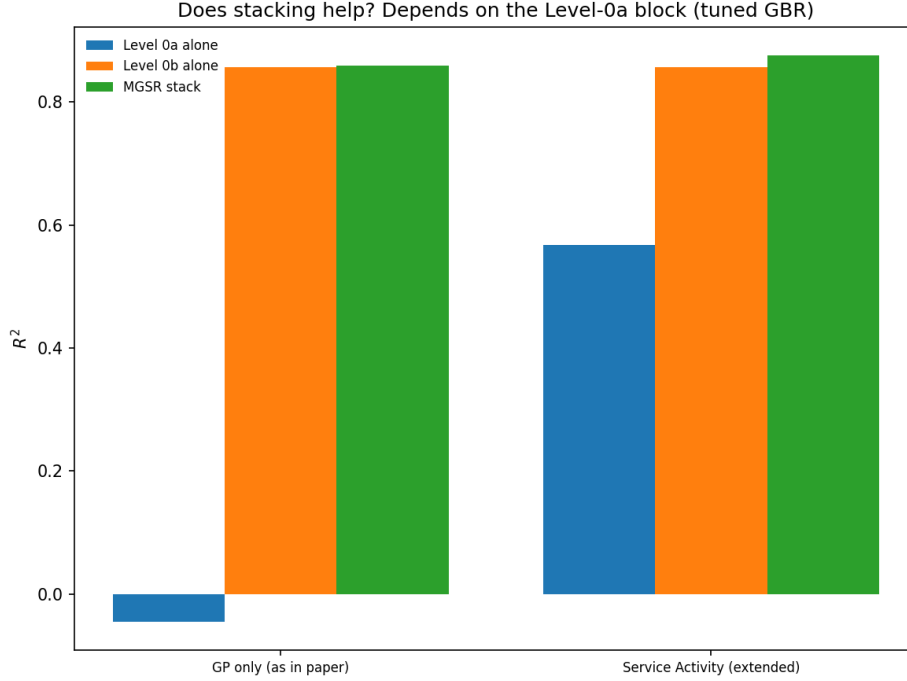


Figure 4.2: Level-0a alone, Level-0b alone, and the stacked model, under the two Level-0a definitions (gradient boosting Level-0 models, repeated 5-fold cross-validation). The stack matches or exceeds its submodels in every configuration, with a genuine margin only where Level-0a carries signal (the extended block).

its own feature block, the flat model wins outright under GBR and ties under RF. The correct conclusion, therefore, is more narrow than the rejection of the architecture: the multi-granular separation preserves the information carried in both feature sets and combines them effectively, but if a sufficiently flexible learner is given access to both sets, then it is able to match or exceed the same performance without the additional structure. The stack’s justification cannot rest on accuracy alone. Its remaining advantages which are interpretable Level-1 weights that quantify the relative contribution of each signal type, and the ability to update submodels independently as data of different vintages arrive are real but are not accuracy gains, and should be argued as such.

Table 4.6: Forecast decomposition: temporal holdout on 2024, gradient boosting. Each row adds one feature block to the specification. The jump from the second to the third row isolates the contribution of the demand lag.

Specification	R^2	MAPE
Static features only (health + demographics)	0.53	13.9%
+ lagged service activity	0.46	15.1%
+ lagged demand (one-month lag)	0.85	7.3%

4.5 Discussion

Generalisation beyond the observed geographies. In the cross-validated values shown above, it is assumed that the model is going to be applied to geographies which have already been seen. However, when the goal is predicting into geographies that have no historical demand, which is what the original paper claims is possible with the cross-sectional methodology, the appropriate test involves withholding geographies completely. Under the GroupKFold cross-validation based on Sub-ICB, the full model falls from $R^2 = 0.91$ to 0.08 (Figure 4.3), while the demographic-only model drops from 0.86 to -0.73 . As would be expected, the failure of the demographic model comes as a result of being just an area indicator, meaning that once it refers to an unobserved geography, it contains no information at all.

Table 4.7 sets the four schemes side by side. Careful interpretation of these figures is required. In group-wise holdout, the baseline global mean is negative (-0.12 by Sub-ICB, -0.21 by ICB) because of

systematic differences in the mean between the held-out groups and the training data, as R^2 is computed based on the variance of the test fold. With respect to the correct baseline, the full model does contribute modest additional information when predicting an unseen Sub-ICB (0.08 against -0.12), but it does not when predicting an unseen ICB (-0.39 against -0.21). There is an immediate explanation for the discrepancy. As the Fingertips indicators have been provided at ICB level, a held-out ICB offers to the model an impossible combination of health features, which tree ensembles are not able to extrapolate from their training distribution. Importantly, the extended service-activity block is close to zero under both schemes (0.03 by Sub-ICB, -0.04 by ICB) and is the only feature set that does not collapse dramatically below the baseline in either. This happens because it includes actual time-dependent quantities instead of area-level constants.

It means that the model can be used to predict Sub-ICBs in the ICBs for which there is existing data, but cannot be transferred to commissioning geographies which have never been observed before, for instance, after the restructuring of the NHS in future.

Table 4.7: The same gradient boosting specification under four validation schemes. Under group-wise holdout the global-mean baseline is itself negative, because held-out groups differ systematically in level from the training data and R^2 is computed against the variance of the test fold.

Validation scheme	Model R^2	Baseline R^2
Repeated 5-fold, random rows	0.91	0.81 (group mean)
Temporal holdout, 2024	0.56	—
GroupKFold by Sub-ICB	0.08	-0.12 (global mean)
GroupKFold by ICB	-0.39	-0.21 (global mean)

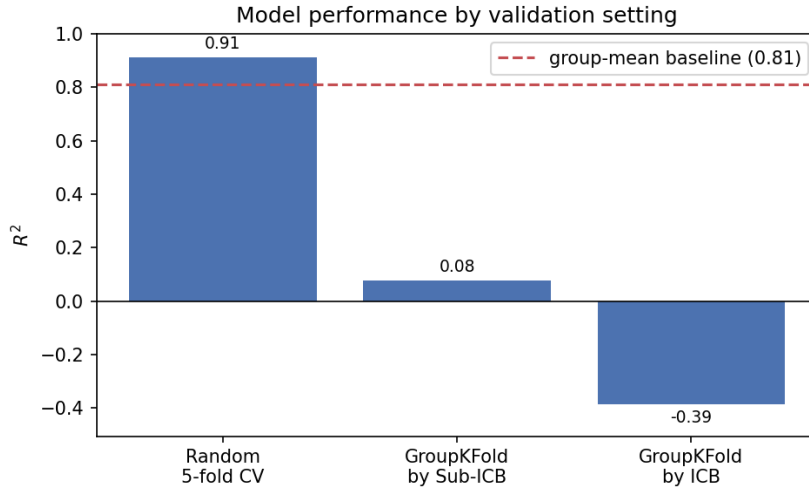


Figure 4.3: The same gradient boosting specification evaluated under progressively stricter validation schemes, with the group-mean baseline shown as a dashed line. The gap between random k -fold and the remaining columns is the share of the cross-validated score attributable to the model having already observed the area.

Forecasting accuracy. For a Random Forest model with a feature set of population health and demographic factors combined with service usage one month ago and demand one month ago trained from April 2022 to December 2023 and tested on the 12 months of 2024, $R^2 = 0.86$ and $\text{MAPE} = 7.2\%$; the corresponding gradient boosting specification is essentially tied at $R^2 = 0.85$, $\text{MAPE} = 7.3\%$. These results outperform the naive persistence method ($R^2 = 0.74$, $\text{MAPE} = 9.3\%$), the seasonal naive method ($R^2 = 0.68$, $\text{MAPE} = 10.9\%$), the Sub-ICB training mean ($R^2 = 0.69$, $\text{MAPE} = 11.4\%$), and both pure time-series approaches estimated separately for each Sub-ICB: exponential smoothing ($R^2 = 0.62$, $\text{MAPE} = 11.2\%$) and SARIMA(1, 1, 1)(1, 0, 0)₁₂ ($R^2 = 0.41$, $\text{MAPE} = 11.1\%$).

The weakness of the time-series models reflects the length of the available series rather than the methods themselves: only 21 monthly observations per Sub-ICB precede the forecast year, and a twelve-period seasonal term estimated from fewer than two complete cycles is barely identified.

Table 4.8 reports every specification and baseline on identical test rows. The decomposition of this result is far more important than the headline itself. The $R^2 = 0.53$ when a specification with only the

static variables of health and demography is used; with the lagged service activity, the score drops to 0.46 under gradient boosting; finally, when the lagged target variable is included, the result jumps to 0.85. In essence, virtually all forecasting power is captured in an autoregressive term alone. This is the result predicted by the literature overviewed in Chapter 2 - Vollmer et al. [25] report the average demand for the previous week as the leading factor across all machine learning models tested; meanwhile, Poursoltan et al. [17] report that a purely econometric specification is within 0–3% of a hybrid machine-learning specification. The multi-granular model was constructed for the environment where there is no demand history and the task is to forecast demand through structural factors. When demand history is available, which is the case after 2022, the flat specification with a single lagged variable performs much better.

This is a final qualifier on the nature of accuracy, where the model excels on the observation-by-observation measure, but the ordering is reversed once the results are aggregated to national monthly averages, because model monthly errors span the range from -6.6% to $+6.4\%$, whereas persistence spans the much narrower range from -3.8% to $+3.9\%$ (Figure 4.4). The model provides a better representation of the demand distribution across Sub-ICBs than persistence does, but persistence represents the temporal pattern of demand at the national level better than the model does. For capacity planning at the national level, persistence is the right tool to use; for capacity planning between commissioning areas, the model is.

Table 4.8: Forecast accuracy for 2024, trained on April 2022 – December 2023. All models and baselines are scored on identical test rows ($n = 900$, 75 Sub-ICBs). Baselines in italics.

Specification	R^2	MAPE
RF: static + activity lag-1 + demand lag-1	0.86	7.2%
GBR: static + activity lag-1 + demand lag-1	0.85	7.3%
<i>Persistence (last 2023 value)</i>	<i>0.74</i>	<i>9.3%</i>
<i>Sub-ICB training mean</i>	<i>0.69</i>	<i>11.4%</i>
<i>Seasonal naive (same month 2023)</i>	<i>0.68</i>	<i>10.9%</i>
Exponential smoothing (per Sub-ICB)	0.62	11.2%
RF: static features only	0.58	12.9%
MGSF (lagged Level-0a + static Level-0b)	0.57	13.7%
GBR: static features only	0.53	13.9%
RF: static + activity lag-1	0.51	14.3%
GBR: static + activity lag-1	0.46	15.1%
SARIMA(1, 1, 1)(1, 0, 0) ₁₂ (per Sub-ICB)	0.41	11.1%

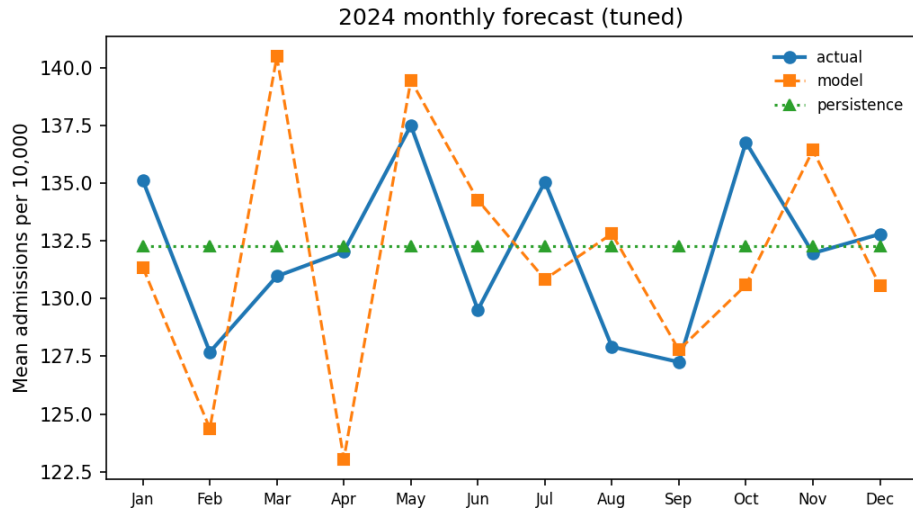


Figure 4.4: Mean monthly emergency admission rate across Sub-ICBs in 2024: actual, model forecast, and persistence. The model tracks the cross-sectional distribution more closely than persistence but reproduces the national monthly profile less closely, which is why the ordering reverses under aggregation.

Limitations. There were four limitations with respect to interpreting the results of the analysis. The first limitation relates to the dependent variable in the models being estimated. The data on Sub-ICB coded emergency department visits for the post-2022 time period is unavailable; thus, admissions are

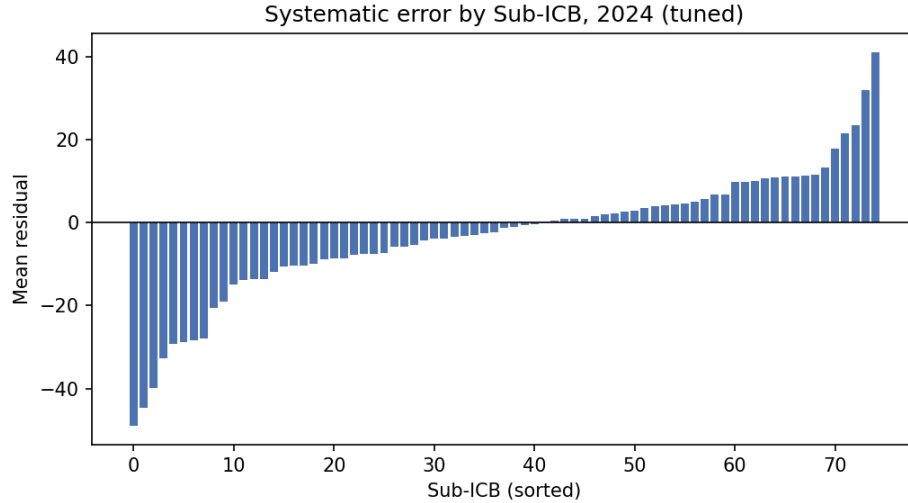


Figure 4.5: Mean 2024 residual by Sub-ICB, sorted. Systematic over- and under-prediction persists at the area level, consistent with the inability of tree ensembles to extrapolate beyond the range of demand observed in training.

used as a proxy for emergency visits. Unlike visit counts, admissions are sensitive to hospital capacity and admission threshold levels besides the demand for emergency care, and the two measures are not equivalent. As a result, the error metrics reported in Chapter 4 cannot be compared with the error metrics reported in Chapter 3.

Secondly, the sample of observations used in this study is not a random sample from the set of all Sub-ICBs in England. The 22 Sub-ICBs excluded because of incomplete Fingertips dataset have significantly smaller means of emergency admission rate than those retained in the sample (112.1 against 130.4 per 10,000, $t = 4.77$, $p < 0.001$); however, there is no difference in terms of population size ($p = 0.35$) and age composition ($p = 0.51$).

Third, the population health model runs at the ICB scale rather than Sub-ICB scale, and cannot be designed such that it differentiates between the Sub-ICBs that share a common ICB parent. The health module’s performance is therefore a benchmark of how well the true Sub-ICB level health measures perform and its inability in a hold-out test is simply due to this restriction. Figure 4.5 shows the resulting error remaining systematic at the area level.

Fourth, the models measure association, but not causation [21] and it does not mean that higher GP appointments lead to high admissions since the correct interpretation would be that both of these variables react to some underlying demand within the population. No *causal* policy conclusions can be drawn from this exercise, though the predictive findings carry practical implications for how demand models of this kind should be deployed. A related limitation is that all models operate at area level; the underlying relationships are individual-level, and the ecological fallacy [19] means that area-level associations need not hold at the level of individual patients.

Chapter 5

Conclusion

This study set out to establish whether the Multi-Granular Stacked Regression of James, Wood and Denholm (2023) survives the structural break that separates the English commissioning landscape of 2018–2019 from that of the post-2022 period, and what its reported accuracy actually measures. Four conclusions follow:

First, the complementary-information hypothesis on which the MGSR rests holds at Sub-ICB level: the stacked model either matches or outperforms both of its submodels in every configuration tested, and the fitted Level-1 weights of 0.27 for the capacity subprediction and 0.84 for population health under the extended specification (0.13 and 0.98 under the capacity block as specified) lie close to the published 0.22 and 0.85. The correspondence survives a change of geography, a change of dependent variable and a wholesale replacement of the population health feature set, which is the clearest positive finding of this work.

Second, the architecture’s justification cannot rest on accuracy because, given a sufficiently flexible learner with access to both feature blocks simultaneously, a single flat model significantly outperforms the stack under tuned gradient boosting ($R^2 = 0.912$ against 0.875, $p = 0.0001$) and is statistically indistinguishable from it under Random Forest (0.852 against 0.863, $p = 0.073$). The multi-granular separation preserves and combines the information in both blocks effectively, but it does not extract information that a unified model cannot, meaning that where the learner is flexible enough, the additional structure becomes a net cost. Its remaining advantages, namely interpretable Level-1 weights and the ability to update submodels independently as data of different vintages arrive, are genuine, but they are architectural rather than predictive and should therefore be argued on those terms.

Third, and most consequentially for the wider literature, the cross-validated figures conventionally reported for models of this kind substantially overstate what has been learned. A baseline that predicts each Sub-ICB using its own historical mean, without any features, already reaches an R^2 of 0.81, while the full sixteen-feature tuned model reaches 0.91, meaning that the genuine incremental contribution of the measured features is only around 0.10 and comes almost entirely from the monthly-varying activity block rather than from the population health features around which the architecture was designed. Under geography-wise holdout, the model retains only modest predictive information for an unseen Sub-ICB ($R^2 = 0.08$) and none for an unseen ICB, suggesting that much of its apparent performance comes from information specific to the Sub-ICBs already observed rather than from patterns that generalise across different geographies. Standard k -fold cross-validation cannot distinguish a model that has learned a transferable relationship from one that has learned which area a row belongs to, and reporting it without a group-mean baseline conceals which of the two has occurred. The practical consequence is that the cross-sectional architecture supports prediction within a panel of observed commissioning areas, but does not support the transfer to areas with no demand history that motivates it.

Fourth, where demand history exists, the multi-granular framing appears to be the wrong one, as a Random Forest model combining structural features with a one-month activity lag and a one-month demand lag forecasts 2024 at $R^2 = 0.86$ and MAPE = 7.2%, outperforming persistence, seasonal naive, exponential smoothing, and SARIMA, while removing the lag term reduces the same specification to $R^2 = 0.58$. This suggests that the complementarity that matters after 2022 is between structural and temporal information rather than between the two blocks of structural information separated by MGSR, meaning that the architecture’s proper domain is the estimation of demand where historical data are unavailable, which is a narrower claim than the original paper makes but a defensible one.

Taken together, these findings carry a practical implication for how models of this kind should be deployed. Where boundary changes have left an area with no usable demand history, as the 2022 NHS reorganisation did and as any future restructuring could do again, the multi-granular architecture is a defensible way to estimate demand from structural features alone. Where an unbroken monthly series does exist, however, the picture is different, with a Random Forest model combining structural features with a one-month activity lag and a one-month demand lag forecasting 2024 best ($R^2 = 0.86$, $\text{MAPE} = 7.2\%$), although naive persistence is not far behind ($R^2 = 0.74$), while both pure time-series methods, SARIMA and exponential smoothing, fall well below it. The seasonal terms on which these methods depend are not reliably found from the 21 pre-forecast months available per Sub-ICB. So, although the tuned tree ensemble is the best choice performance-wise, its advantage over persistence is small enough that it is worth checking its added complexity against a persistence benchmark for any given series rather than assuming its superiority. Commissioners reading such figures should therefore treat a single cross-validated R^2 with caution and ask for the group-mean baseline alongside it, since if the model barely improves on that baseline, much of its apparent accuracy may come from learning which area each row belongs to rather than from a relationship that would generalise to an area it has never seen, which is precisely the use case a structural forecasting tool is intended to serve.

Two extensions follow directly from these findings: The first is a working-days covariate, which would isolate the residual within-area association between elective and emergency activity from the common calendar driver acting on both. The second is a longer panel as the series are too short for seasonal time-series methods to be identified properly at 33 months and two more years of Sub-ICB data would provide enough history for these methods to be compared more fairly with the cross-sectional model.

Appendix A

Data Sources and Reproducibility

This appendix lists all data sources used in the replication (Chapter 3) and extension (Chapter 4), including publication names, geographic resolution, date ranges and coverage. All resources are publicly available without registration. The analysis code and derived datasets for the panels are available at [GitHub](#).

A.1 Replication Data Sources (2018–2019)

Table A.1: Data sources for the 2018–2019 CCG-level replication.

Variable	Publication	Published level	Date range	Coverage
GP appointments	NHS Digital, Appointments in General Practice	CCG	Oct 2018 – Dec 2019	73 CCGs; Jan–Sep 2018 filled from authors’ file
Ambulance calls	NHS England, AmbSYS Time Series	Ambulance trust (11)	Jan 2018 – Dec 2019	64 CCGs independent; 9 from authors’ file
NHS 111 calls	NHS England, NHS 111 Minimum Dataset	111 area (53)	Jan 2018 – Dec 2019	48 CCGs independent; 25 from authors’ file
Population	ONS Mid-Year Population Estimates (SAPE21, SAPE22)	CCG (E38 codes)	2018, 2019	64 CCGs independent; 9 from authors’ file
People, Places, Lives	ONS Health Index	Local authority	2018, 2019	73 CCGs via LSOA crosswalk
ED attendances	NHS England, A&E Attendances and Emergency Admissions	Trust	Jan 2018 – Dec 2019	Used authors’ published values (trust-to-CCG mapping not available)

Table A.2: Data sources for the 2022–2024 Sub-ICB-level extension.

Variable	Publication	Published level	Date range	Coverage
Emergency admissions (target)	NHS Digital, HES Monthly Activity Report	Sub-ICB (patient-coded)	Apr 2022 – Dec 2024	97 Sub-ICBs; 75 in final sample
Elective admissions, outpatient attendances, outpatient DNAs, GP referrals	NHS Digital, HES Monthly Activity Report	Sub-ICB	Apr 2022 – Dec 2024	Same file as target
GP appointments	NHS Digital, Appointments in General Practice	Sub-ICB	Apr 2022 – Dec 2024	93.9% row coverage
QOF prevalence (asthma, COPD, CHD, depression, diabetes, heart failure, hypertension)	OHID Fingertips platform via <code>fingertips_py</code>	ICB	2022, 2023, 2024	77.3% row coverage; mapped to Sub-ICBs via parent ICB
Population and age structure	ONS Mid-Year Sub-ICB Population Estimates	Sub-ICB	2022, 2023, 2024	Full coverage for retained Sub-ICBs

A.2 Extension Data Sources (2022–2024)

A.3 Sample Construction

The replication sample consists of 1,435 CCG-month observations across 73 CCGs, compared with 1,618 observations across 74 CCGs in the original study. The difference arises from slightly different treatment of missing values during the merge.

The extension sample was constructed as follows:

1. Starting panel: 3,298 observations (97 Sub-ICBs $\times$ 34 months, with one duplicate month).
2. Removal of 97 duplicated November 2022 rows (byte-identical records from overlapping HES annual files): 3,201 observations.
3. Restriction to complete cases across all feature sets (Fingertips QOF at 77.3% coverage, GP appointments at 93.9%): 2,325 observations across 75 Sub-ICBs in 25 ICBs.
4. The 22 excluded Sub-ICBs have significantly lower mean emergency admission rates (112.1 vs. 130.4 per 10,000; $t = 4.77$, $p < 0.001$) but do not differ in population size ($p = 0.35$) or age composition ($p = 0.51$).

All models, baselines and significance tests are evaluated on this common sample with identical cross-validation folds.

A.4 Software and Reproducibility

All modelling was conducted in Python 3 using `scikit-learn` for Random Forest, Gradient Boosting, cross-validation and evaluation metrics; `xgboost` for the XGBoost specifications; `statsmodels` for SARIMA and exponential smoothing baselines; and `pandas/NumPy` for data processing. All random seeds are set to 0 (`random_state=0`) throughout. Hyperparameters for each model role are reported in Table 4.1. Cross-validation folds were generated once and reused across all specifications to permit paired Wilcoxon signed-rank tests.

A.4.1 Data accessibility statement

All data used in this study are publicly available. Emergency admissions and secondary-care activity were obtained from the NHS Digital Hospital Episode Statistics Monthly Activity Report; GP appointment counts from NHS Digital Appointments in General Practice; chronic disease prevalence from the Office for Health Improvement and Disparities Fingertips platform via the `fingertips_py` API client [18]; and population and age structure from ONS mid-year population estimates. The replication dataset for 2018–2019 was reconstructed from NHS England AmbSYS and NHS 111 Minimum Dataset publications, ONS mid-year estimates and the ONS Health Index. The code and the derived panel are available at [Github Repository](#).

Bibliography

- [1] Ronald M. Andersen. Revisiting the behavioral model and access to medical care: Does it matter? *Journal of Health and Social Behavior*, 36(1):1–10, 1995.
- [2] James Bergstra and Yoshua Bengio. Random search for hyper-parameter optimization. *Journal of Machine Learning Research*, 13:281–305, 2012.
- [3] Leo Breiman. Random Forests. *Machine Learning*, 45(1):5–32, October 2001.
- [4] Tianqi Chen and Carlos Guestrin. XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794. ACM, 2016.
- [5] Qingyu Cheng, Nilay Tanik Argon, Christopher S. Evans, Yufeng Liu, Timothy F. Platts-Mills, and Serhan Ziya. Forecasting emergency department hourly occupancy using time series analysis. *American Journal of Emergency Medicine*, 48:177–182, 2021.
- [6] Thomas E. Cowling, Matthew J. Harris, Hilary C. Watt, Daniel C. Gibbons, and Azeem Majeed. Access to general practice and visits to accident and emergency departments in England: cross-sectional analysis of a national patient survey. *British Journal of General Practice*, 64(624):e434–e439, July 2014.
- [7] Jerome H. Friedman. Greedy function approximation: A gradient boosting machine. *The Annals of Statistics*, 29(5):1189–1232, October 2001.
- [8] Méliッサ Fritsch, Cyrielle Brossard, Mahuna Akplogan, Pauline Bouquet, Camille Dib, Amandine Schnee, Christophe Goetz, and Laure Abensur Vuillaume. Impact of sporting, cultural and social events on emergency admissions: a retrospective multicenter study. *BMC Emergency Medicine*, 25(1):217, October 2025.
- [9] Sally A. Hull, Kate Homer, Kambiz Boomla, John Robson, and Mark Ashworth. Population and patient factors affecting emergency department attendance in London: retrospective cohort analysis of linked primary and secondary care records. *The British Journal of General Practice: The Journal of the Royal College of General Practitioners*, 68(668):e157–e167, March 2018.
- [10] Rob J. Hyndman and George Athanasopoulos. *Forecasting: Principles and Practice*. OTexts, 3 edition, 2021.
- [11] Charlotte James, Richard Wood, and Rachel Denholm. A multi-granular stacked regression for forecasting long-term demand in Emergency Departments. *BMC Medical Informatics and Decision Making*, 23:29, February 2023.
- [12] Shancheng Jiang, Qize Liu, and Beichen Ding. A systematic review of the modelling of patient arrivals in emergency departments. *Quantitative Imaging in Medicine and Surgery*, 13(3):1957–1971, March 2023.
- [13] Evangelos Kontopantelis, David A Springate, Mark Ashworth, Roger T Webb, Iain E Buchan, and Tim Doran. Investigating the relationship between quality of primary care and premature mortality in England: a spatial whole-population study. *The BMJ*, 350:h904, March 2015.
- [14] NHS England. A&e attendances and emergency admissions. <https://www.england.nhs.uk/statistics/statistical-work-areas/ae-waiting-times-and-activity/>, 2026. NHS England Statistics, accessed September 2026.

- [15] NHS England. Integrated care in your area, 2026. Page updated periodically; ICB boundaries as at April 2026.
- [16] Office for National Statistics. Health index for england: 2015 to 2021, 2023.
- [17] Lily Poursoltan, Jie Cao, Brian Clay, Beth Trimble, Leah Adrid, Jeffrey Pan, Andrew Chua, John Bell, Christopher A. Longhurst, Kevin Zhu, and Karandeep Singh. Prospective comparison of econometric, machine learning, and foundation models for forecasting emergency department boarding patients. *npj Health Systems*, 2(1):49, December 2025.
- [18] Public Health England. fingertips_py: Python package for the fingertips api. https://github.com/publichealthengland/PHDS_fingertips_py, 2026. Accessed 2026.
- [19] W. S. Robinson. Ecological correlations and the behavior of individuals. *American Sociological Review*, 15(3):351–357, 1950.
- [20] Bahman Rostami-Tabar and Florian Ziel. Anticipating special events in emergency department forecasting. *International Journal of Forecasting*, 38(3):1197–1213, September 2022.
- [21] Galit Shmueli. To Explain or to Predict? *Statistical Science*, 25(3):289–310, August 2010.
- [22] Eduardo Silva, Margarida F. Pereira, Joana T. Vieira, João Ferreira-Coimbra, Mariana Henriques, and Nuno F. Rodrigues. Predicting hospital emergency department visits accurately: A systematic review. *The International Journal of Health Planning and Management*, 38(4):904–917, July 2023.
- [23] Vidya K. Sudarshan, Mikkel Brabrand, Troels M. Range, and Uffe Kock Wiil. Performance evaluation of emergency department patient arrivals forecasting models by including meteorological and calendar information: A comparative study. *Computers in Biology and Medicine*, 135:104541, 2021.
- [24] The King’s Fund. What’s going on with A&E waiting times?, 7 2024.
- [25] Michaela A.C. Vollmer, Ben Glampson, Thomas Mellan, Swapnil Mishra, Luca Mercuri, Ceire Costello, Robert Klaber, Graham Cooke, Seth Flaxman, and Samir Bhatt. A unified machine learning approach to time series forecasting applied to demand at emergency departments. *BMC Emergency Medicine*, 21(1):9, January 2021.
- [26] M. Wargon, B. Guidet, T. D. Hoang, and G. Hejblum. A systematic review of models for forecasting the number of emergency department visits. *Emergency medicine journal: EMJ*, 26(6):395–399, June 2009.
- [27] David H. Wolpert. Stacked generalization. *Neural Networks*, 5(2):241–259, 1992.
- [28] Youri Yordanov, Agathe Beauvais, and Pierre-Clément Thiébaud. Multimorbidity in emergency departments: urgent need for integrated care. *BMJ medicine*, 3(1):e000989, 2024.